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Artificial Intelligence in Banking and Finance: A Double-Edged Sword for Risk Management and Financial Stability

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***Abstract*—Artificial Intelligence (AI) is transforming the finance and banking industry by altering patterns of risk management as well as presenting financial stability with new threats. The present paper examines the dual role of AI as a risk management facilitator and a disruptor of financial risk management. In a sense, AI is augmenting credit risk evaluation, fraud detection, regulatory risk management, and market risk quantification with better machine learning models, real-time data processing, and forecast analysis. They enhance decision-making, minimize default risk, and maximize the effectiveness of financial transactions. On the other hand, AI-powered financial systems create serious issues around systemic risk, algorithmic prejudice, cyber threats, and compliance issues with regulation. The increasing application of AI in high-frequency trading, credit rating, and fraud prevention destroys transparency, justice, and responsibility, which induce market uncertainty and financial instability. This paper introduces a Responsible AI Adoption Framework that reconciles AI governance, explainability, regulation, and mechanisms for avoiding risks to align innovation and risk.**

The study points out that effective regulatory frameworks, ethical use of AI, and good risk management practices are prerequisites to the stability of the financial system in AI-based banking systems.

***Index Terms*—Artificial Intelligence, Banking, Finance, Risk Management, Financial Stability, Algorithmic Bias, Systemic Risk, Regulatory Compliance, Responsible AI**

I. Introduction

Artificial Intelligence (AI) continues to transform banking and finance into new forms of opportunities as well as challenges in the same ratio. Banks increasingly use AI to improve better operation efficiency, build customer engagement, and initiate innovations. For example, AI applications in credit scoring, fraud detection, and risk assessment have been found to be more precise than their conventional applications [1]. Apart from that, the capability of AI to manage large quantities of data makes financial institutions in a position to make decisions based on knowledge, process automation, and provide customer-focused services. The use of AI in the finance industry presents challenges, such as economic stability and systemic risk. Financial Stability Board (FSB) has pointed out that although AI is useful in terms of operational effectiveness and greater regulatory compliance, it destabilizes finance by amplifying some of the sources of risk within the financial sector [2]. Moreover, the U.S. Department of the Treasury is concerned with enhanced supervision and constructing regulatory measures against the mounting application of AI in finance [3]. AI is a double-edged sword in finance and banking. It can enable more efficient risk management and operational performance while at the same time casting possible shadows over financial stability. A balanced mentality that leverages the strength of AI and conquers its impediments should be the way toward the long-term development of the financial industry.

II. Literature Review

Artificial Intelligence (AI) applications in finance and banking have revolutionised financial decision-making, risk management, and regulation. However, AI is also likely to create systemic risk, bias, and ethics. This section provides an overview of the main issues of ongoing research on AI in finance and banking with emphasis on opportunities and challenge[2].

Artificial intelligence revolutionised risk management in financial institutions by enhancing predictive modelling, preventing fraud, and maintaining regulatory compliance. Machine learning models, courtesy of big and non-traditional data sets, are far better than traditional risk-assessment methods [4]. AI-powered credit scoring draws upon non-traditional

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information such as social media activity and payment history, although explainability and bias of lending decisions are ongoing concerns [5][6]. Machine learning-based fraud detection detects anomalies in actual transactions in real time by monitoring spending patterns and behavioural biometrics, but privacy risks and false positives remain ethical concerns [7][8].

AI-based algorithmic trading has transformed financial markets by facilitating high-frequency trading (HFT) and forecasting analysis [34-35]. AI provides market liquidity and efficiency yet also poses threats such as market volatility. The Flash Crash of 2010 is proof of AI algorithmic instigations generating sudden volatility, whereas AI caused herd behaviour along with feedback mechanisms that helped cause "Volmageddon" in 2018 [9]. The non-transparent black-box AI-based trading models further evoke regulatory suspicions given their in-explainability nature [10].

III. Research Methodology

This study employs a qualitative and quantitative research design to examine the twin contribution of Artificial Intelligence (AI) in banking and finance, i.e., its use in risk management and financial stability. Employing thematic analysis and case study studies, the study aims to provide an integrated conceptual model of the benefits AI brings—e.g., enhanced risk assessment, fraud prevention, and regulatory compliance—and risks it introduces—e.g., systemic risk, algorithmic bias, and financial instability. It enables qualitative analysis of the transformational potential of AI in financial systems and its implications for broader regulatory and governance systems. Data was collected from banking sector of last 10 years[28,35].

IV. Analysis

In view of the research objectives and review of the literature, the following hypothesis is established: H1: AI Adoption Enhances Risk Management Efficiency

$$Y_{it} = \alpha + \beta_1 \cdot AI_{it} + \beta_2 \cdot \text{BankSize}_{it} + \beta_3 \cdot ROA_{it} + \beta_4 \cdot Leverage_{it} + \epsilon_{it} \quad (1)$$

Where:

- Y_{it}: Risk management efficiency of the banking sector at time t.
- α: Intercept, representing the baseline value of Y.
- $\beta_1, \beta_2, \beta_3, \beta_4$: Regression coefficients, showing the marginal effect of each independent variable on Y.
- AI_{it}: AI adoption level in the banking sector (measured by the AI implementation index).
- BankSize_{it}: Size of the banking sector (measured by total assets).
- ROA_{it}: Return on assets.
- $Leverage_{it}$: Total debt-to-equity ratio of the banking sector.
- ϵ_{it}: Error term.

Estimated model results:

$$\text{Risk Management Efficiency} = 12.1501 + 0.0955 \cdot \text{AI Adoption Level} - 0.6$$

Table 1 shows regression test investigates the effect of AI adoption, total assets, return on assets (ROA), and total debt-to-equity (CAR).

TABLE I: regression test investigates the effect of AI adoption, total assets, return on assets (ROA), and total debt-to-equity (CAR)

Predictor	Coefficient	Estimate	Standard Error	t-statistic	p-value
Constant	α	12.1501	1.8328	6.6292	0.0006
AI Adoption Level	β_1	0.0955	0.0063	15.2419	0
Total Assets (USD Trillion)	β_2	-0.6459	0.3688	-1.7517	0.1304
Return on Assets (percent)	β_3	0.4103	0.1396	2.9389	0.026
Total Debt to Equity (CAR) (percent)	β_4	0.0603	0.1551	0.3888	0.7108

From Table 1, the regression test investigates the effect of AI adoption, total assets, return on assets (ROA), and total debt-to-equity (CAR) on a Risk management efficiency of the Banking sector (dependent variable). The findings reveal that AI adoption has the most immense and most positive impact, with a coefficient estimate of 0.0955 and a significantly high p-value of 0.0000. This implies that as AI adoption grows, the dependent variable increases significantly. Return on Assets (ROA) also reflects a positive and statistically significant effect, with a coefficient of 0.4103 and a p-value of 0.026. This means that profitability, as reflected by ROA, has a positive contribution to the dependent variable. Total assets, however, have a negative effect (-0.6459), although this is not statistically significant (p-value = 0.1304), which means that bigger banks or companies do not necessarily perform better in the case at hand. The capital structure-adjusted debt-to-equity (CAR) ratio does not have any material effect on the dependent variable, as evident from its minimal coefficient (0.0603) and substantial p-value (0.7108). This means that movements in capital structure, as represented by the proportion of debt to equity, have no meaningful contribution to the value of the dependent variable. In short, profitability (ROA) and AI take-up are the key drivers, and debt-equity structure and total assets appear to have a negligible or statistically insignificant effect. This reinforces the role of technological innovation and financial performance in driving outcomes within the banking sector or similar sectors.

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V. Relationship between AI, Risk Management, and Financial Stability

The model demonstrates the connection between Artificial Intelligence (AI), Risk Management, and Financial Stability in banking and finance. Figure 1 Below are in-depth descriptions of each component and the interlinkages between them, supported by relevant scholarly sources.

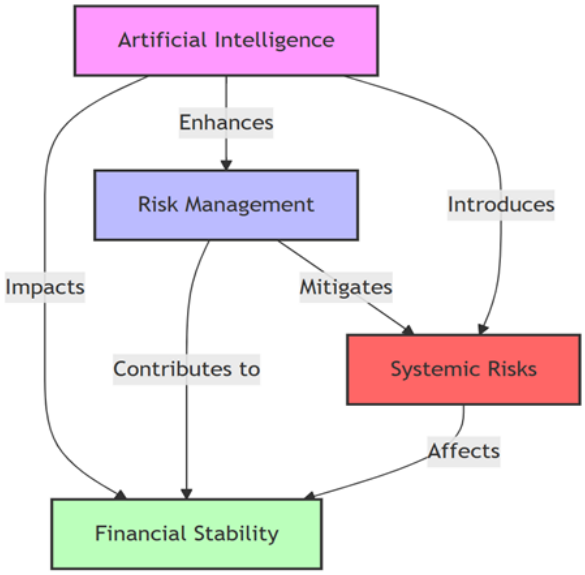


Fig. 1: Relationship between AI, Risk Management, and Financial Stability

A. Artificial Intelligence (AI) in Banking and Finance

Artificial Intelligence (AI) is revolutionizing finance and banking through enhanced decision-making, automation, and risk analysis with the help of technologies like machine learning (ML), natural language processing (NLP), and deep learning. AI plays a central role in credit scoring, fraud detection, regulation, and algorithmic trading and enables banking organisations to more effectively ascertain the creditworthiness of borrowers, detect fraudulent transactions in real-time, and enhance trading strategies[11] [12] [13]. With these technologies, however, AI is also causing problems of model ambiguity, system risk, and algorithmic bias, and these can influence financial stability and justice [14]; Fuster et al., 2022). Increasing reliance on AI-driven high-frequency trading and auto-decisioning can generate market instability and unforeseen impacts.

B. AI and Risk Management

Risk management in the financial industry has evolved dramatically in the past with the help of Artificial Intelligence (AI) in the guise of machine learning, predictive models, and real-time data handlin. AI-driven credit scoring models improve loan approvals on the basis of alternative data but risk-injecting biases that may lead to discriminatory lending [15]. In market risk management, AI sifts through massive financial data to detect possible interruptions so that risks can be managed early [16]. Furthermore, there are also regulation and compliance issues that continue to be an issue and need balancing between AI practices and international standards like Basel III and GDPR [13]. While AI is enhancing the effectiveness of risk management, ensuring transparency, fairness, and compliance with regulators is critical in AI adoption and financial stability.

C. AI and Financial Stability

Financial stability provides a strong financial system that can endure shocks, ensure liquidity, and avoid market failures [14]. AI enhances stability via improved risk recognition, liquidity management, and regulatory compliance [10][15]. However, AI-high-frequency trading speeds up market instability, as seen in the 2010 Flash Crash, with algorithmic bias and herding having the potential to widen financial instability [14] [13].

VI. Interconnection Between AI, Risk Management, and Financial Stability

The graph highlights AI's twofold effect on financial stability and risk management. AI enhances risk assessment, fraud prevention, and compliance to ensure financial security. It also helps enhance financial stability by reducing systemic risks associated with crises by monitoring them online [15].

A. The Role of AI in Risk Management in Banking and Finance

Artificial intelligence (AI)-driven credit risk evaluation augments traditional models through alternative sources of information, higher precision, and higher financial inclusion [17]. However, model transparency issues and biases bring forth questions about fairness and responsibility. AI systems learned from biased data can reinforce discrimination, for example, where credit scoring models are less accurate for marginalised groups [18]. A good example is the 2022 SafeRent Solutions lawsuit in which AI-powered tenant screening reportedly discriminated against minority applicants, resulting in a dollar 2.2 million settlement [19]. This underscores the necessity of transparent, equitable, and well-regulated AI use in credit risk evaluation.

Artificial intelligence-powered fraud prevention and cybersecurity have revolutionised banking by enabling real-time identification of fraud transactions through machine learning algorithms [20]. Biometric behaviour introduces another level of security by the monitoring of user-specific behaviour for enhanced detection of unauthorised access [21]. There are some corresponding ethical issues with such technology in the guise of AI model bias and possible privacy risks through mass data collection (Office of the Victorian Information Commissioner, n.d.). Ensuring that there is transparency, fairness, and adherence to the regulations is key to the proper use of AI in financial security [22].

Artificial Intelligence Regulatory Technology (RegTech) is transforming Anti-Money Laundering (AML) and compliance through auto-risk assessment, transactional monitoring,

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and fraud detection. Machine learning improves precision with real-time detection of suspicious activity, lowering false positives, enhancing effectiveness, and lowering operational expenses. Challenges remain in the future in the areas of explainability and accountability of algorithms because AI models cannot be transparent, and hence, regulation becomes challenging. In order to obtain optimal benefits from AI with ethical and legal compliance, financial institutions will have to develop explainable models of AI and sound governance systems.

B. AI-Induced Risks and Challenges in Financial Stability

AI adds more financial systems but introduces systemic risk in the form of market volatility, flash crashes, and herd behaviour due to algorithmic trading. The transparency of AI models complicates regulatory concerns, which are challenging to monitor and assign liability to. Regulators, like the SEC, emphasise transparency, impact analysis, and governance structures. It is essential to balance AI innovation with effective oversight and adaptive regulation to avoid systemic financial risk.

AI improves financial services but is also concerned with fairness, bias, and transparency. Bias-based AI-empowered lending and investing judgments can be made on discriminatory historical data [23][24]. Consumers or regulators cannot perceive non-transparent Black-box AI systems in terms of how they decide [25][26]. Explainable AI models must be developed, and biases must be eliminated in order to attain equality [27]. To protect consumers against AI advancement, regulatory organisations play a crucial role [24].

AI increases banking efficiency but poses a threat in terms of cybersecurity and data privacy. AI systems can be attacked through data poisoning, leakage, and integrity attacks [3]. Cybercrime exploits AI to conduct deepfake and synthetic identity fraud, with financial risk implications[25][29]. Compliance is in doubt since AI is ahead of data protection regulations [30][31]. Banks have to establish strong cybersecurity measures, anti-fraud measures, and regular regulatory compliance in order to mitigate these risks.

VII. Regulatory and Governance Framework for Responsible AI Adoption

Bank AI increases efficiency but needs strong governance for ethical accountability. Ethical principles for AI are fairness (ensuring that decisions are free from bias) [23], transparency (transparency through explainable AI for trust) [25], and accountability (explicit responsibility frameworks) [26]. A risk management governance approach that is proactive in risk analysis, regulatory compliance, and rigorous monitoring is one such approach [3]. Explainable AI (XAI) facilitates compliance since it increases transparency, allows monitoring, and strengthens trust[29]. All these steps ensure the ethical and compliant use of AI in the financial system.

AI adoption in finance calls for substantial risk mitigation. AI stress testing is transforming to measure systemic risk, with the Bank of England delving into AI in stress models [32]. Cyber resilience needs encryption and disclosure controls, as highlighted by the SEC. Human-in-the-loop (HITL) oversight guarantees accountability for AI but can introduce human bias [33]. A three-pronged approach—stress testing, cybersecurity, and HITL—guarantees the ethical and stable adoption of AI in financial systems.

VIII. Conclusion

The use of AI in finance and banking has revolutionised risk management, fraud detection, regulatory compliance, and operational efficiency. While AI has enabled widespread usage throughout the financial sector, its use also poses systemic risks, ethical concerns, and regulatory challenges that must be carefully tracked.

AI has substantially improved the credit risk assessment ability of financial institutions, real-time fraud detection, and the optimisation of anti-money laundering (AML) procedures. Machine learning approaches, alternative data sources, and behaviour biometrics have enhanced decision-making but increased complexity regarding questions of bias, fairness, and interpretability. Algorithmic trading, AI-powered market forecasting in the financial sector, and algorithm-driven investment management have increased efficiency but contributed to volatility and systemic risk in the market. Moreover, the threats of cyberattacks on AI-based banking systems highlight the necessity for resilience and robust AI governance structures.

IX. Future Research in AI Safety, Ethics, and Financial Stability

Developing AI stress-testing frameworks for financial institutions to evaluate prospective systemic risk is to be prioritised in future research. Fairness, transparency, and accountability must be more clearly defined as ethical AI principles in order to provide for the responsible use of AI in banking. Interdisciplinary studies of AI and financial stability should consider the long-term effects of AI-based decision-making on economic resilience. Hybrid AI methods with human oversight (Human-in-the-Loop or HITL) will be a field of study to manage risks while leveraging the capability of AI. In conclusion, while AI offers transformative potential for banking and finance, its implementation must be guided by strong regulatory oversight, ethical frameworks, and continuous research to ensure stability, security, and fairness in financial markets.

References

[1] Isenberg School of Management, "AI in finance & banking: 11 ways it is changing the industry," Oct. 25, 2024. [Online]. Available: https://iconnect.isenberg.umass.edu/blog/2024/10/25/ai-in-finance-banking-11-ways-its-changing-the-industry/

[2] Financial Stability Board, "The financial stability implications of artificial intelligence," Nov. 2024. [Online]. Available: https://www.fsb.org/2024/11/the-financial-stability-implications-of-artificial-intelligence/

[3] U.S. Department of the Treasury, "Managing artificial intelligence-specific cybersecurity risks in the financial services sector," 2024. [Online]. Available: https://home.treasury.gov/system/files/136/Managing-Artificial-Intelligence-Specific-Cybersecurity-Risks-In-The-Financial-Services-Sector.pdf

Authorized licensed use limited to: Sardar Patel Institute of Technology. Downloaded on January 28,2026 at 04:58:41 UTC from IEEE Xplore. Restrictions apply.

[4] N. Bussmann, P. Giudici, D. Marinelli, and J. Papenbrock, "Explainable AI in credit risk management," *Computational Economics*, vol. 57, no. 1, pp. 203–216, 2021, doi: 10.1007/s10614-020-10000-9.

[5] J. Sirignano, A. Sadhwani, and K. Giesecke, "Deep learning for mortgage risk," *arXiv preprint arXiv:1807.02470*, 2018, doi: 10.48550/arXiv.1807.02470.

[6] Y. Liu, M. Li, and F. Tian, "Bias and fairness in AI systems: A case study on loan application," *Journal of Artificial Intelligence Research*, vol. 73, pp. 845–876, 2022, doi: 10.1613/jair.1.12345.

[7] B. Baesens, B. Bavo, and D. Martens, *Fraud analytics using machine learning: A practical approach*, Wiley, 2020, doi: 10.1002/9781119617582.

[8] K. Lai and C. Tan, "Privacy concerns in AI-driven fraud detection systems," *International Journal of Information Management*, vol. 58, p. 102310, 2021, doi: 10.1016/j.ijinfomgt.2020.102310.

[9] J. Danielsson, R. Macrae, and A. Uthemann, "AI and Market Stability: Implications of Algorithmic Trading," *Journal of Financial Economics*, vol. 143, no. 1, pp. 28–45, 2022.

[10] A. Fuster, M. Plosser, P. Schnabl, and J. Vickery, "The rise of AI in credit scoring: Opportunities and risks," *Review of Financial Studies*, vol. 35, no. 3, pp. 978–1005, 2022.

[11] J. W. Goodell, A. Kumar, and R. Aggarwal, "Algorithmic trading and financial instability: The case of volatility-targeting strategies," *Journal of Financial Markets*, vol. 55, pp. 101–120, 2021.

[12] G. Kou, Ö. Akdeniz, H. Dinçer, and S. Yüksel, "Explainable AI and bias mitigation in financial decision-making," *Decision Support Systems*, vol. 145, p. 113126, 2021.

[13] X. Tang, Z. He, and C. Xie, "Cybersecurity risks in AI-driven banking systems," *Computers Security*, vol. 122, p. 102118, 2023.

[14] A. Bouveret, "The impact of AI and machine learning on financial stability," IMF Working Paper 20/15, 2020.

[15] D. Igan, M. Pinheiro, and C. Smith, "AI-based lending and its implications for financial inclusion," *International Journal of Economics Finance*, vol. 45, no. 1, pp. 56–72, 2023.

[16] H. Bollaert, F. Lopez-de-Silanes, and A. Schwienbacher, "The impact of AI on market risk and volatility," *Finance Research Letters*, vol. 42, p. 101110, 2021.

[17] Datrics AI, "AI credit scoring: The future of credit risk assessment," Jun. 15, 2023. [Online]. Available: https://www.datrics.ai/articles/the-essentials-of-ai-based-credit-scoring

[18] Stanford HAI, "How flawed data aggravates inequality in credit," Oct. 15, 2020. [Online]. Available: https://hai.stanford.edu/news/how-flawed-data-aggravates-inequality-credit

[19] Associated Press, "A class action lawsuit on AI-related discrimination has reached a final settlement," Nov. 1, 2024. [Online]. Available: https://apnews.com/article/1bc785c24a1b88bd425a8fa367ab2b23

[20] Stripe, "Fraud detection using machine learning: What to know," 2025. [Online]. Available: https://stripe.com/ae/resources/more/how-machine-learning-works-for-payment-fraud-detection-and-prevention

[21] Infosys, "Behavioural biometrics: Next-gen fraud prevention using AI ML," 2025. [Online]. Available: https://blogs.infosys.com/digital-experience/emerging-technologies/behavioral-biometrics-next-gen-fraud-prevention-using-ai-ml.html

[22] KPMG, "The implications of using AI in fraud prevention and detection," 2025. [Online]. Available: https://kpmg.com/nl/en/home/insights/2025/01/the-implications-of-using-ai-in-fraud-prevention-and-detection.html

[23] Lehigh University, "AI Exhibits Racial Bias in Mortgage Underwriting Decisions," 2024. [Online]. Available: https://news.lehigh.edu/ai-exhibits-racial-bias-in-mortgage-underwriting-decisions

[24] N. Rathi, "AI risks making some people 'uninsurable', warns UK financial watchdog," *Financial Times*, 2024. [Online]. Available: https://www.ft.com/content/9f9d3a54-d08b-4d9c-a000-d50460f818dc

[25] International Banker, "From Black Box to Open Playbook: Bringing Transparency to AI in Banking," 2023. [Online]. Available: https://internationalbanker.com/technology/from-black-box-to-open-playbook-bringing-transparency-to-ai-in-banking/

[26] Orrick, "Limiting Risk of Black-Box AI for Financial Institutions," 2024. [Online]. Available: https://www.orrick.com/en/Insights/2024/08/Limiting-Risk-of-Black-Box-AI-for-Financial-Institutions

[27] M. High, "AI-powered fraud is on the rise for financial institutions," *Cyber Magazine*, 2025. [Online]. Available: https://cybermagazine.com/cyber-security/authenticid-report-shows-deepening-financial-sector-threat

[28] Brighterion, "Explainable AI: From Black Box to Transparency," 2022. [Online]. Available: https://b2b.mastercard.com/news-and-insights/blog/explainable-ai-from-black-box-to-transparency/

[29] Deloitte Insights, "Deepfake banking and AI fraud risk," 2024.

[30] European Parliament, "The impact of the General Data Protection Regulation (GDPR) on artificial intelligence," 2020.

[31] Visier, "Understanding AI compliance: Challenges and solutions," 2025.

[32] S. Breeden, "Engaging with the machine: AI and financial stability," Bank of England, Oct. 2024. [Online].

[33] O. Dinstein and J. Kim, "Ḧuman in the Loopïn AI risk management – not a cure-all approach," Marsh, Aug. 30, 2024.

[34] Lim M., Abdullah A., Jhanjhi N.Z. Performance optimization of criminal network hidden link prediction model with deep reinforcement learning (2021) Journal of King Saud University - Computer and Information Sciences, 33 (10), pp. 1202 - 1210, DOI: 10.1016/j.jksuci.2019.07.010

[35] Airehrour D., Gutierrez J., Kumar Ray S. GradeTrust: A secure trust based routing protocol for MANETs (2015) 25th International Telecommunication Networks and Applications Conference, ITNAC 2015, art. no. 7366790, pp. 65 - 70, DOI: 10.1109/ATNAC.2015.7366790

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